

A+ Core 1 Memorization Sheet

Print this and keep it beside you. Core 1 is a recall exam: port numbers, wireless standards, cable categories, connector names, RAM and drive facts, the laser printing steps, cloud vocabulary, and which symptom points to which part. Every line here is fair game.

01 The exam itself

FACT	VALUE
Exam	CompTIA A+ Core 1, 220-1201 (A+ needs Core 1 and Core 2, 220-1202)
Questions and time	Up to 90 questions in 90 minutes; multiple choice plus performance-based items
Passing	675 on a scale of 100 to 900 (Core 2 needs 700)
Hardware and Network Troubleshooting	28%
Hardware	25%
Networking	23%
Mobile Devices	13%
Virtualization and Cloud Computing	11%

Troubleshooting and hardware are more than half the exam. For every part: what it is called, its number (speed, pins, volts, distance), how it fails, and what you check first.

02 Ports and protocols

PORT	PROTOCOL	TRANSPORT	PURPOSE
20, 21	FTP	TCP	File transfer; 21 commands, 20 data; clear text
22	SSH	TCP	Secure remote shell; also SFTP and SCP
23	Telnet	TCP	Clear-text remote terminal; replaced by SSH
25	SMTP	TCP	Sending mail between servers (587 for client submission with TLS)
53	DNS	UDP and TCP	Name to address lookups (TCP for large answers and zone transfers)
67, 68	DHCP	UDP	Automatic addressing; server listens on 67, client on 68
80	HTTP	TCP	Web, unencrypted
110	POP3	TCP	Download mail to one device (995 secure)
137 to 139	NetBIOS / NetBT	UDP 137, 138; TCP 139	Legacy Windows name service and file sharing
143	IMAP	TCP	Mail kept on the server, synced across devices (993 secure)
161, 162	SNMP	UDP	Device monitoring; 162 traps
389	LDAP	TCP	Directory queries (636 LDAPS)
443	HTTPS	TCP	Web over TLS
445	SMB / CIFS	TCP	Windows file and printer sharing
3389	RDP	TCP	Remote Desktop
123	NTP	UDP	Time synchronization

PORT	PROTOCOL	TRANSPORT	PURPOSE
514	Syslog	UDP	Log collection
5900	VNC	TCP	Cross-platform screen sharing

TCP versus UDP

- TCP: connection-oriented (three-way handshake), reliable, ordered, acknowledged, retransmits; web, mail, file transfer, remote desktop
- UDP: connectionless, no acknowledgment, low overhead; DNS lookups, DHCP, streaming, VoIP, SNMP, syslog, TFTP
- A port identifies the service at an address; well-known ports are 0 to 1023; a firewall filters by address and port

03 Wireless

STANDARD	WI-FI NAME	BAND	TOP SPEED	NOTES
802.11a		5 GHz	54 Mb/s	Old; short range
802.11b		2.4 GHz	11 Mb/s	Oldest common; DSSS
802.11g		2.4 GHz	54 Mb/s	Backward compatible with b
802.11n	Wi-Fi 4	2.4 and 5 GHz	600 Mb/s	MIMO, 40 MHz channels
802.11ac	Wi-Fi 5	5 GHz only	About 6.9 Gb/s	MU-MIMO, 80 and 160 MHz channels
802.11ax	Wi-Fi 6 and 6E	2.4 and 5 GHz; 6E adds 6 GHz	About 9.6 Gb/s	OFDMA, better in crowds
802.11be	Wi-Fi 7	2.4, 5, and 6 GHz	About 46 Gb/s	320 MHz channels, multi-link

BAND	RANGE AND PENETRATION	CHANNELS	CROWDING
2.4 GHz	Longest range, best through walls	1 to 11 in the US; only 1, 6, and 11 do not overlap at 20 MHz	Microwaves, Bluetooth, cordless phones, neighbors
5 GHz	Shorter range	Many non-overlapping channels; some require DFS (radar avoidance)	Less crowded; 20, 40, 80, 160 MHz widths
6 GHz	Shortest range	Widest clean spectrum; Wi-Fi 6E and 7 only	Newest devices only

Other radios

- Channel width: wider channels carry more data but overlap more; regulations set which channels and power levels a country allows
- Bluetooth: 2.4 GHz personal area network; pairing with a PIN; class 2 about 10 m; low energy for wearables
- NFC: 13.56 MHz, about 4 cm, tap to pay and pair; RFID: tags read by radio at a distance, passive (no battery) or active
- Long-range fixed wireless and cellular (3G, 4G LTE, 5G) serve as internet connections where cable and fiber are absent

04 Network hardware and services

DEVICE	WHAT IT DOES	REMEMBER
Router	Connects networks; forwards by IP address; the default gateway	Layer 3; NAT and DHCP in SOHO routers
Switch	Connects devices in one network; forwards by MAC address per port	Layer 2; managed (VLANs, PoE settings, monitoring) versus unmanaged (plug and play)
Access point	Bridges wireless clients to the wired network	Not a router by itself; SSID, channel, security

DEVICE	WHAT IT DOES	REMEMBER
Patch panel	Terminates building cable runs at the closet; patch cables to switch ports	Punched down with a 110 tool; labeling
Firewall	Allows or blocks traffic by address, port, and rule	Stateful; hardware or software
PoE injector / PoE switch	Sends power over the data cable to APs, phones, cameras	802.3af 15.4 W; 802.3at PoE+ 30 W; 802.3bt PoE++ 60 W (Type 3) and 100 W (Type 4)
Cable modem	Internet over the cable TV coax (DOCSIS)	F-type connector; shared neighborhood bandwidth
DSL modem	Internet over the phone line	RJ11; distance to the exchange limits speed
ONT	Optical network terminal; converts fiber to Ethernet at the premises	Fiber to the home
NIC	Network interface card; wired or wireless	Burned-in 48-bit MAC address, six hex pairs, first half is the manufacturer
Hub	Obsolete repeater that sends every frame to every port	Collisions; replaced by switches

SERVER ROLE OR APPLIANCE	PROVIDES
DNS	Names to addresses
DHCP	Addresses, masks, gateways, DNS servers by lease
File share and print server	Shared folders (SMB) and shared printers with queues
Mail server	SMTP out, IMAP or POP3 in
Syslog	Collects logs from devices
Web server	HTTP and HTTPS
AAA	Authentication, authorization, accounting (RADIUS, TACACS+)
Database server	SQL data for applications
NTP	Time
Spam gateway	Filters inbound mail before the mail server
UTM	Unified threat management: firewall, IDS/IPS, antivirus, content filter in one box
Load balancer	Spreads requests across several servers; high availability
Proxy server	Fetches web content on behalf of clients; caching, filtering, logging
SCADA and legacy embedded	Industrial control; old, fragile, isolate on their own network
IoT	Smart devices: thermostats, cameras, speakers, locks; segregate and patch

05 IP addressing and configuration

ITEM	VALUE
IPv4 address	32 bits, four octets 0 to 255, such as 192.168.1.20
Private ranges	10.0.0.0 to 10.255.255.255 (/8); 172.16.0.0 to 172.31.255.255 (/12); 192.168.0.0 to 192.168.255.255 (/16)
Public addresses	Everything else; routable on the internet; shared through NAT at the router
APIPA	169.254.x.x: the client got no DHCP answer; link or DHCP server problem
Loopback	127.0.0.1 (IPv6 ::1): tests the local TCP/IP stack

ITEM	VALUE
Subnet mask	255.255.255.0 = /24 (254 hosts); 255.255.0.0 = /16; 255.0.0.0 = /8; same network portion means direct delivery, otherwise via the gateway
Default gateway	The router's address on the local network; must be in the same subnet as the host
Static versus dynamic	Static typed in (servers, printers); dynamic from DHCP (workstations, phones); reservations give a fixed address by MAC
IPv6 address	128 bits, eight hex groups; :: compresses zeros; fe80:: link-local on every interface; 2000::/3 global; no NAT needed

DNS RECORD	PURPOSE
A	Name to IPv4 address
AAAA	Name to IPv6 address
CNAME	Alias of another name
MX	Mail server for a domain
TXT	Free text; carries SPF, DKIM, and DMARC
SPF (TXT)	Which servers may send mail for the domain
DKIM (TXT)	Cryptographic signature proving the message was not altered and came from the domain
DMARC (TXT)	Policy telling receivers what to do when SPF or DKIM fails (none, quarantine, reject) and where to send reports

DHCP, VLAN, VPN

- DHCP scope: the pool of addresses; lease: how long a client keeps one; reservation: a fixed address tied to a MAC; exclusion: addresses in the range never handed out
- DORA: Discover, Offer, Request, Acknowledge
- VLAN: a logical network on a managed switch; separates traffic (voice, guest, devices) without separate hardware; a router connects VLANs
- VPN: an encrypted tunnel across a public network; site-to-site or client; the client appears to be on the office network

06 Internet connection and network types

CONNECTION	MEDIUM	TRAITS
Fiber	Glass, ONT at the premises	Fastest, symmetric, immune to interference
Cable	Coax, DOCSIS modem	Fast, shared with neighbors, asymmetric
DSL	Copper phone line	Distance-limited, slower; always-on
Satellite	Dish to orbit	Available anywhere; high latency; weather affects it; data caps
Cellular	4G LTE, 5G	Mobile; hotspot and tethering; caps
WISP	Fixed wireless from a tower to an antenna	Rural; line of sight

NETWORK TYPE	SCOPE
PAN	Personal: Bluetooth, a few meters
LAN	One building or site
WLAN	A LAN over Wi-Fi
MAN	A city or campus
WAN	Between sites, across regions; the internet
SAN	Storage area network: block storage over a dedicated network (Fibre Channel, iSCSI)

07 Cables and connectors

COPPER CATEGORY	SPEED	BANDWIDTH	NOTES
Cat 5	100 Mb/s	100 MHz	Obsolete
Cat 5e	1 Gb/s	100 MHz	Minimum for gigabit
Cat 6	1 Gb/s (10 Gb/s to 55 m)	250 MHz	Tighter twists, spline
Cat 6a	10 Gb/s to 100 m	500 MHz	Common for 10 gigabit
Cat 7	10 Gb/s	600 MHz	Shielded; GG45 or RJ45
Cat 8	25 or 40 Gb/s to 30 m	2,000 MHz	Data center runs

Copper rules

- Maximum run 100 m (328 ft) including patch cables; RJ45 eight-pin plug; RJ11 four or six-pin phone plug
- T568A order: white-green, green, white-orange, blue, white-blue, orange, white-brown, brown. T568B: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown
- Same standard both ends = straight-through (device to switch); A on one end and B on the other = crossover (like devices; rarely needed with auto-MDIX)
- UTP unshielded (normal); STP shielded for high-interference areas; plenum-rated jackets for air-handling spaces (low smoke); direct burial for outdoors
- Coax: RG-6 for cable TV and modems, F-type screw connector; RG-59 for short CCTV runs; BNC bayonet connector

FIBER	CORE	SOURCE	DISTANCE	CONNECTORS
Single-mode	About 9 microns	Laser	Kilometers	ST (bayonet twist), SC (square push-pull), LC (small clip, most common)
Multimode	50 or 62.5 microns	LED or VCSEL	Hundreds of meters (OM3 about 300 m at 10 Gb/s)	Same connectors; orange or aqua jackets

INTERFACE	SPEED	NOTES
USB 2.0	480 Mb/s	5 m cable; black port; Type A, B, mini, micro
USB 3.0 (3.2 Gen 1)	5 Gb/s	3 m; blue port; SuperSpeed
USB 3.1 Gen 2	10 Gb/s	Teal or red port
USB 3.2 Gen 2x2	20 Gb/s	USB-C only
USB4	40 Gb/s (80 in version 2)	USB-C; tunnels Thunderbolt, DisplayPort, PCIe
Thunderbolt 3 and 4	40 Gb/s	USB-C connector; daisy chain; power delivery; Thunderbolt 5 is 80 Gb/s
USB-C	Connector, not a speed	Reversible 24-pin; can carry USB, DisplayPort alt mode, Thunderbolt, and up to 240 W power delivery
Lightning	USB 2.0 speeds	Apple 8-pin reversible; replaced by USB-C on newer devices
Serial RS-232	Kilobits	DB9 connector; console ports on network gear
SATA	1.5, 3, 6 Gb/s (SATA I, II, III)	7-pin data, 15-pin power; 1 m cable; eSATA external to 2 m
Molex	Power only	4-pin 5 V and 12 V for older drives and fans

VIDEO	SIGNAL	NOTES
VGA	Analog	15-pin DE-15, blue; degrades with length; no audio
DVI	Digital (DVI-D), analog (DVI-A), both (DVI-I)	Single link 1920 x 1200; dual link 2560 x 1600; no audio
HDMI	Digital video and audio	19-pin; Type A full, C mini, D micro; HDMI 2.1 up to 48 Gb/s, 8K; CEC control
DisplayPort	Digital video and audio	20-pin with a latch; multi-stream daisy chaining; DP 2.0 up to 80 Gb/s; mini DisplayPort
USB-C DisplayPort alt mode	Digital	One cable for video, data, and power

08 Displays

TYPE	TRAITS
LCD TN (twisted nematic)	Cheapest, fastest response, poor viewing angles and color
LCD IPS (in-plane switching)	Best color and viewing angles; slower and pricier; professional and most laptops
LCD VA (vertical alignment)	High contrast, deep blacks; middle on angles and speed
OLED	Each pixel emits its own light; perfect blacks, thin, flexible; risk of burn-in; used in phones and premium laptops
Mini-LED	An LCD with thousands of tiny backlight zones; high brightness and contrast without burn-in
Inverter	Converts DC to AC for CCFL backlights in old LCDs; a dim screen visible with a flashlight means backlight or inverter
Touch screen and digitizer	The digitizer layer senses touch or pen; a cracked digitizer works badly even if the image is fine

Attributes

- Resolution: 1920 x 1080 Full HD; 2560 x 1440 QHD; 3840 x 2160 4K UHD; the native resolution is the only one that looks sharp
- Pixel density in pixels per inch: the same resolution on a smaller panel is sharper
- Refresh rate in hertz: 60 standard, 120 to 240 for gaming; the cable and port must support the rate at the resolution
- Color gamut: sRGB (web), Adobe RGB and DCI-P3 (print and video); a wider gamut shows more colors

09 Memory, storage, and RAID

RAM	PINS (DIMM / SODIMM)	VOLTAGE	TYPICAL SPEEDS	NOTES
DDR3	240 / 204	1.5 V (1.35 V low voltage)	800 to 2133 MT/s	Keyed differently from DDR4
DDR4	288 / 260	1.2 V	2133 to 3200 MT/s	Most common in service now
DDR5	288 / 262	1.1 V	4800 MT/s and up	Different key from DDR4; on-module power management; two channels per module

Memory rules

- Generations are not interchangeable; the notch position enforces it. SODIMM for laptops and small form factors; DIMM for desktops
- ECC RAM detects and corrects single-bit errors; servers and workstations; the board and CPU must support it. Non-ECC for consumer systems

- Channels: install matched modules in the color-coded paired slots for dual (or quad) channel bandwidth; a single module runs single channel
- Virtual memory (paging file) uses disk when RAM is full; constant paging means add RAM

STORAGE	FACTS
HDD	Spinning platters; 5,400, 7,200, 10,000, 15,000 rpm; 2.5-inch (laptop) and 3.5-inch (desktop); slow, cheap per TB; fails with clicking and grinding
SSD SATA	No moving parts; about 550 MB/s limited by SATA III; 2.5-inch or mSATA or M.2 SATA
SSD NVMe	Talks PCIe directly; M.2 or add-in card; thousands of MB/s (Gen 3 about 3.5 GB/s, Gen 4 about 7 GB/s)
M.2	A slot and form factor, not an interface: keys B, M, or B+M; sizes 2242, 2260, 2280 (22 mm wide, length in mm); can be SATA or NVMe
SAS	Serial Attached SCSI; enterprise drives, 12 Gb/s, dual ports; SAS controllers accept SATA drives, not the reverse
Removable	USB flash drives; SD, microSD, CompactFlash memory cards; speed classes on the card
Optical	CD 700 MB; DVD 4.7 GB (8.5 dual layer); Blu-ray 25 GB (50 dual layer)

RAID	HOW	MINIMUM DRIVES	SURVIVES	USABLE CAPACITY
0	Striping	2	No drive failure	100% (sum of drives)
1	Mirroring	2	One drive	50%
5	Striping with distributed parity	3	One drive	All but one drive
6	Striping with double parity	4	Two drives	All but two drives
10 (1+0)	Stripe of mirrors	4	One drive per mirrored pair	50%

10 Motherboards, CPUs, firmware, and power

FORM FACTOR	SIZE	NOTES
ATX	12 x 9.6 in (305 x 244 mm)	Full size; most expansion slots
microATX	9.6 x 9.6 in	Fits ATX cases; fewer slots
Mini-ITX	6.7 x 6.7 in (170 mm)	Small builds; one expansion slot

CONNECTOR	FACTS
PCIe	Lanes x1, x4, x8, x16; a card fits in a longer slot; per lane about 1 GB/s (Gen 3), 2 GB/s (Gen 4), 4 GB/s (Gen 5) each direction; graphics cards use x16
PCI	Legacy 32-bit parallel slot; 133 MB/s shared
Main power	24-pin (20+4) ATX; 4- or 8-pin EPS for the CPU; 6-pin (75 W) and 8-pin (150 W) PCIe for graphics; 12VHPWR 16-pin up to 600 W
Drive power	SATA 15-pin; Molex 4-pin
SATA and eSATA	7-pin data ports on the board; eSATA on the rear panel
Headers	Front panel (power, reset, LEDs), USB 2.0 and 3.0, audio, fan (3-pin voltage, 4-pin PWM), RGB
M.2	Key and length printed by the slot; some slots are SATA only, some NVMe only, some both
CPU sockets	Intel LGA (pins in the socket, such as LGA 1700 and 1851); AMD AM4 PGA (pins on the chip) and AM5 LGA; board chipset must match the CPU generation; servers may be multisocket

BIOS / UEFI SETTING	WHY IT MATTERS
UEFI versus legacy BIOS	UEFI: GPT disks over 2 TB, Secure Boot, graphical setup, network boot; legacy uses MBR
Boot options	Order of devices; USB or network boot for installs; disable unused to harden
USB permissions	Disable ports or block storage devices to prevent data theft
TPM	Trusted Platform Module: stores keys for BitLocker and Windows Hello; TPM 2.0 required by Windows 11; can be firmware (fTPM) or a chip
Secure Boot	Only signed boot loaders run; blocks boot-sector malware; may need disabling for some Linux or older media
Passwords	Supervisor (setup) versus user (boot) password; cleared by the CMOS jumper or battery removal
Fans and temperature	Fan curves, temperature monitoring, thermal shutdown thresholds
Virtualization support	Intel VT-x or AMD-V must be enabled for hypervisors
HSM	Hardware security module: a dedicated external device for key storage and crypto in servers and enterprises
CMOS battery	CR2032 coin cell; a dead one resets time, date, and settings at every boot

CPUs and cooling

- x86 is 32-bit, x64 is 64-bit (more than 4 GB of RAM); ARM is a low-power RISC design in phones, tablets, Apple silicon, and some laptops; software must be built for the architecture
- Cores run separate work at once; simultaneous multithreading (Hyper-Threading) shows two threads per core; cache L1, L2, L3 speeds up access; clock in GHz
- Expansion cards: video (GPU, VRAM), sound, capture (recording HDMI input), NIC (wired or wireless); need a free slot, power, and a driver
- Cooling: case fans (intake front, exhaust rear), CPU heat sink and fan, thermal paste or pad between CPU and sink (replace when reseating), liquid all-in-one coolers; dust is the enemy

POWER SUPPLY	FACTS
Input	110 to 120 VAC (North America, 60 Hz) or 220 to 240 VAC (much of the world, 50 Hz); auto-switching or a red voltage selector; wrong setting destroys the supply
Output rails	3.3 V (orange), 5 V (red), 12 V (yellow), ground (black); 12 V feeds CPU, GPU, and drives
Connectors	24-pin (20+4) main, EPS CPU, PCIe, SATA power, Molex
Wattage	Add the components' draw and leave headroom; a GPU may need 300 W or more by itself
Efficiency	80 Plus Bronze, Silver, Gold, Platinum, Titanium; higher wastes less as heat
Modular	Only the cables you need; better airflow
Redundant	Two supplies in servers; one fails, the other carries the load; hot swappable

11 Printers

LASER STEP	WHAT HAPPENS
1 Processing	The page is rasterized into a bitmap in the printer's memory
2 Charging	The primary charge roller (or corona wire) puts a uniform negative charge (about minus 600 V) on the drum
3 Exposing	The laser writes the image, removing charge where toner should stick
4 Developing	Negatively charged toner from the developer roller clings to the exposed areas
5 Transferring	The transfer roller (or corona) gives the paper a positive charge that pulls toner off the drum
6 Fusing	Heat (about 180 C) and pressure melt the toner into the paper; the fuser is hot

LASER STEP	WHAT HAPPENS	
7 Cleaning	A blade scrapes leftover toner off the drum; a lamp erases the charge	

PRINTER	KEY PARTS	MAINTENANCE
Laser	Imaging drum, toner cartridge, fuser, transfer belt or roller, pickup rollers, separation pad, duplexing assembly	Replace toner; apply the maintenance kit (fuser, rollers, pads) at the page count; calibrate; clean with a toner vacuum, never compressed air
Inkjet	Ink cartridges, print head, carriage belt, feed rollers, duplexer	Clean and align print heads; replace cartridges; calibrate; clear jams; keep heads from drying
Thermal	Heating element, feed assembly, special heat-sensitive paper, no ink	Replace paper; clean the heating element with isopropyl alcohol; remove debris; keep paper away from heat
Impact (dot matrix)	Print head with pins, ribbon, tractor feed, multipart carbonless paper	Replace ribbon, print head, and paper; the only type that prints multipart forms
3D	Filament (FDM) or resin, print bed, nozzle	Level the bed; clear the nozzle; store filament dry

Deployment and settings

- Unbox, remove shipping tape and locks, place near power and network away from heat and dust; update firmware; install the right driver for the OS
- PCL: fast, printer-dependent, everyday output. PostScript: device-independent page description, exact and consistent, used in graphics and publishing
- Connectivity: USB (one PC), Ethernet, wireless; sharing: a printer shared from a PC versus a print server that manages queues and drivers centrally
- Settings: duplex (both sides), orientation (portrait, landscape), tray selection and paper size, quality and draft mode
- Security: user authentication, badge release, audit logs, secured (held) prints released with a PIN at the device
- Scan services: scan to email (SMTP settings), scan to folder (SMB share and credentials), scan to cloud; ADF for stacks, flatbed for books and cards

12 Mobile devices

LAPTOP PART	REPLACEMENT NOTES
Battery	Lithium-ion; internal on most modern laptops; discharge and disconnect before other work; swollen means replace now
Keyboard and keys	Ribbon cable under the palm rest; individual keycaps on scissor or butterfly mechanisms
RAM	SODIMM; many thin laptops solder it (not upgradable)
Storage	2.5-inch SATA or M.2 (SATA or NVMe); check the key and length
Wireless card	M.2 (2230) card; the two antenna leads (main and aux) route through the hinge to the lid; black and white or gray leads must go to their marked terminals
Camera and microphone	In the lid bezel; cable through the hinge; a physical privacy shutter on some
Biometrics and NFC	Fingerprint reader in the power button or palm rest; smart card or NFC reader for badges

Connections and accessories

- USB-C (reversible, data, video, power), microUSB and miniUSB (older), Lightning (Apple, older); NFC tap; Bluetooth pairing; tethering and hotspot share a phone's cellular data
- Docking station: a full set of ports and power with one connection (Thunderbolt or USB-C, or proprietary). Port replicator: adds ports only, usually USB
- Accessories: stylus (active pens need pairing or a battery), headsets, speakers, webcams; trackpad, drawing pad, track point

- Cellular: 3G, 4G LTE, 5G; SIM card or eSIM profile identifies the subscriber; airplane mode; data roaming
- Bluetooth pairing: enable Bluetooth, enable pairing (discoverable), find the device, enter the PIN, test the connection
- Location: GPS satellites (needs sky) and cellular or Wi-Fi positioning (works indoors, less precise)
- MDM: corporate versus BYOD device configurations, policy enforcement (passcodes, encryption, remote wipe), corporate application deployment
- Synchronization: mail, calendar, contacts, cloud storage, business apps; watch data caps on cellular; sync over Wi-Fi when possible

13 Virtualization and cloud

TERM	MEANING
Type 1 hypervisor	Bare metal, runs directly on hardware (server virtualization); best performance
Type 2 hypervisor	Runs as an application on a host OS (desktop virtualization, testing)
Sandbox	An isolated VM to open suspicious files or test changes safely
Test and development	Many OS versions on one machine; snapshots to roll back
Application virtualization	Run legacy software or another OS's programs; cross-platform virtualization
VDI	Virtual desktop infrastructure: desktops run on servers, users connect with thin clients
Containers	Share the host kernel; lighter than VMs; package an application with its dependencies
VM requirements	CPU virtualization enabled, enough RAM and storage for each guest, network mode (bridged, NAT, host-only), and security (patch guests, isolate, guard against VM escape)
CLOUD MODEL	MEANING
Public	Provider's shared infrastructure, pay as you go
Private	Dedicated to one organization, on premises or hosted
Hybrid	Private plus public, workloads move between them
Community	Shared by organizations with common needs (government, healthcare)
IaaS	Rent virtual machines, storage, and networks; you manage the OS up
PaaS	A platform to build and run applications; the provider manages OS and runtime
SaaS	A finished application over the web; email, office suites, CRM

Cloud characteristics

- Shared resources (multitenancy, cheaper) versus dedicated resources (isolation, predictable performance)
- Metered utilization: pay for what you use; ingress (data in) is usually free, egress (data out) is billed
- Elasticity: capacity grows and shrinks with demand automatically; availability: uptime commitments through redundancy
- File synchronization: the same files on every device and in the cloud; conflicts and deleted-file propagation are the traps

14 Troubleshooting

The method (CompTIA's best practice)

- 1 Identify the problem: gather information, question the user, identify changes, back up before making changes. 2 Establish a theory of probable cause; question the obvious; research. 3 Test the theory. 4 Establish a plan of action and implement it. 5 Verify full functionality and implement preventive measures. 6 Document findings, actions, and outcomes.

BOARD, RAM, CPU, POWER SYMPTOM	LIKELY CAUSE AND FIRST CHECK
No power, no fans	Outlet, cord, PSU switch and voltage selector, PSU (test with a tester or paperclip), front panel header
Fans spin, no POST, beeps	Beep codes point to RAM or video; reseal RAM and card; one module at a time
Blank screen after POST	Monitor input, cable, GPU seating and power; onboard video
Random shutdown, overheating, burning smell	Dust, failed fan, dried thermal paste, failing PSU; check temperatures in firmware
Proprietary crash screen (BSOD, pinwheel)	Driver, RAM (run a memory test), storage, overheating; note the stop code
Sluggish performance, application crashes	RAM shortage, thermal throttling, failing drive, malware
Unusual noise	Fan bearing (whine), HDD (clicking, grinding), coil whine under load
Capacitor swelling or leaking	Board or PSU failure; replace
Inaccurate date and time	CMOS battery (CR2032)
DRIVE AND RAID SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Clicking, grinding	HDD mechanical failure; back up immediately, replace
Bootable device not found	Boot order, loose cable, dead drive, corrupted boot record, USB stick left in
S.M.A.R.T. failure warning	Drive predicts its own failure; back up and replace
Extended read and write times, low IOPS	Failing drive, fragmentation on HDD, nearly full SSD, wrong SATA mode
Data loss or corruption	Failing drive, bad cable, power loss during writes; run chkdsk after a backup
RAID failure, array missing, audible alarm	A failed member; replace the drive and rebuild; check the controller and its battery; a missing array may be a controller or driver
Missing drives in the OS	Not initialized or partitioned, driver, cable, power, disabled port in firmware
LED status indicators	Amber or red on a drive bay means degraded or failed
VIDEO AND DISPLAY SYMPTOM	LIKELY CAUSE AND FIRST CHECK
No image, incorrect input source	Select the right input; check the cable and adapter; try another cable
Dim image	Brightness, power saving, failing backlight or inverter; shine a flashlight at the screen
Fuzzy image, sizing issues	Non-native resolution, scaling, VGA over a long cable, projector focus
Flashing or flickering screen	Loose cable, refresh rate, driver, failing panel or inverter
Incorrect color, distorted image	Bent VGA pins, cable, driver, color profile, degaussing on CRTs
Dead pixels	Panel defect; warranty policy; stuck pixels may recover
Burn-in	Static image on OLED or plasma; pixel shift and screensavers prevent it
Burnt-out bulb, intermittent projector shutdown	Replace the lamp (hours counter); overheating from a clogged filter
Audio issues over HDMI or DisplayPort	Select the display as the output device; cable supports audio
MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Poor battery health, short life	Aged battery (check health percentage), background apps, brightness, radios; replace
Swollen battery	Stop using it; do not puncture; replace and dispose properly
Improper charging	Cable, adapter wattage, dirty or damaged port, wrong charger
Overheating	Charging while gaming, direct sun, malware, a failing battery

MOBILE SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Liquid damage	Power off, do not charge, dry thoroughly; indicators inside show exposure
Digitizer issues, cursor drift, touch calibration	Cracked digitizer, screen protector, calibration utility, moisture
Poor or no connectivity	Airplane mode, Wi-Fi or cellular toggles, SIM seating, forget and rejoin the network, carrier settings
Malware, cannot install apps, degraded performance	Unofficial app sources, full storage, OS out of date, factory reset as last resort
Stylus does not work	Battery or pairing on active pens, wrong pen model, digitizer
Broken screen, damaged port	Replace the assembly; ports are soldered on many devices

NETWORK SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Intermittent wireless	Weak signal, interference, channel overlap, roaming between APs, driver
Slow speeds	Congestion, wrong cable category, duplex mismatch, bandwidth hogs, a hub
Limited connectivity, APIPA address	No DHCP: link, cable, VLAN, DHCP server
Intermittent internet	ISP, modem, router overheating, DNS; test the modem directly
Jitter, high latency, poor VoIP	Congestion; no QoS; wireless for voice; satellite links
Port flapping	Link going up and down: bad cable or connector, failing NIC, switch port, duplex or speed mismatch
External interference	Microwaves, motors, fluorescent ballasts, neighbors' Wi-Fi; move to 5 GHz or shielded cable
Authentication failures	Wrong password or key, expired certificate, RADIUS server, MAC filtering, time skew

PRINTER SYMPTOM	LIKELY CAUSE AND FIRST CHECK
Lines or streaks down the page	Dirty or scratched drum, low toner, dirty print head (inkjet), a scratch on the fuser
Faded prints	Low toner or ink, economy mode, transfer roller worn, humidity
Speckling	Loose toner in the printer; leaking cartridge; clean
Double or echo images (ghosting)	Drum not cleaned, worn drum, fuser problem
Toner not fused, smears when touched	Fuser failed or not reaching temperature; wrong paper
Garbled print	Wrong driver or language (PCL versus PostScript), bad cable, corrupted job, low printer memory
Paper jams, multipage misfeed	Worn pickup rollers or separation pad, damp or wrong paper, overfilled tray, debris
Paper not feeding, tray not recognized	Rollers, tray seating, paper guides, tray sensor, firmware
Multiple prints pending, frozen queue	Restart the print spooler service, clear the queue, check the printer status and connectivity
Incorrect page orientation or size	Driver settings, application settings, tray paper size mismatch
Grinding noise	Gears, rollers, drum drive, foreign object; maintenance kit
Finishing issues (staple jams, hole punch)	Clear the finisher, refill staples, check the finisher tray sensors
Connectivity issues	IP change on DHCP (use a reservation), Wi-Fi, cable, sleep mode, firewall

15 Networking and hardware tools

TOOL	USE
Crimper	Attaches RJ45 or RJ11 plugs to cable ends

TOOL	USE
Cable stripper	Removes the jacket without nicking conductors
Punchdown tool	Seats wires into a 110 or 66 block or a keystone jack and trims them
Cable tester	Checks wire map, opens, shorts, split pairs; certifiers measure to category
Toner probe (tone generator and probe)	Finds the far end of a cable in a bundle or closet
Loopback plug	Tests a NIC or switch port by looping transmit to receive
Wi-Fi analyzer	Shows signal strength, channels, and interference; site surveys
Network tap	Copies traffic on a link for a protocol analyzer without disrupting it
Multimeter	Voltage, continuity, resistance; PSU rails and cables
Power supply tester	Checks each PSU rail and the power-good signal
POST card	Reads the firmware's progress codes when a board will not boot
ESD strap and mat	Prevents static damage while handling boards, RAM, and cards